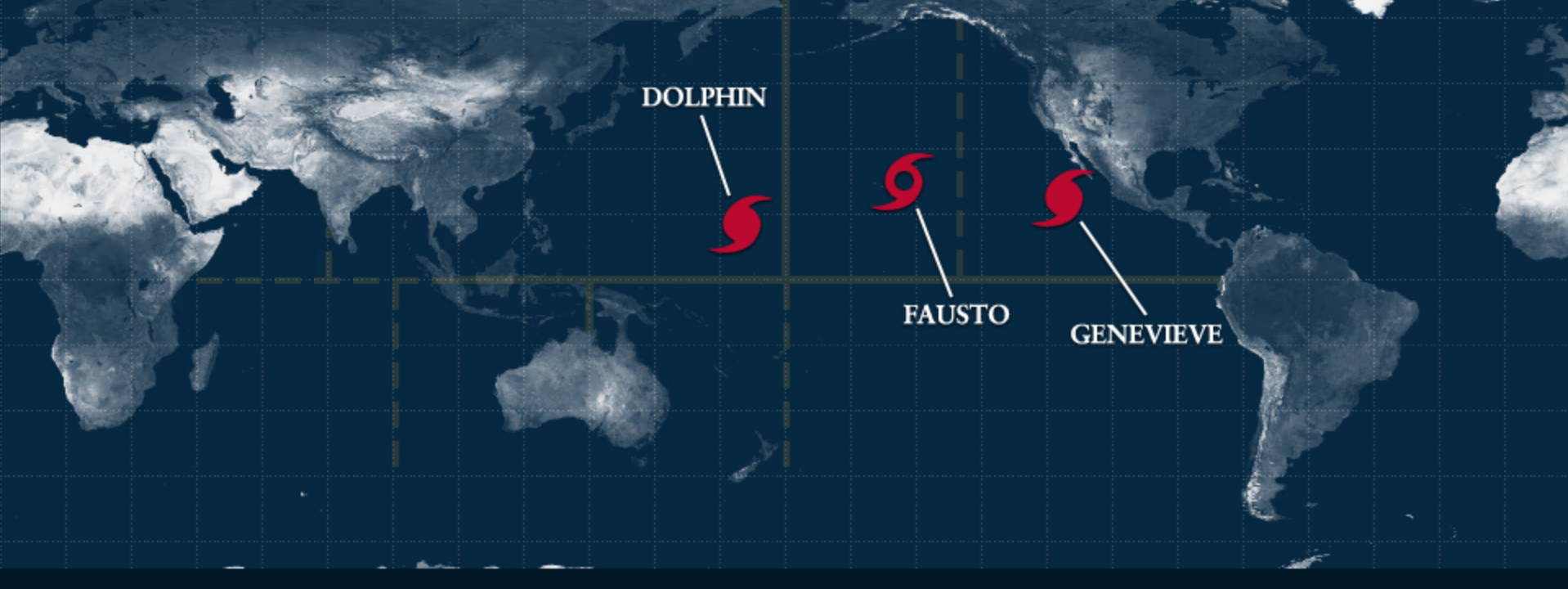


Home Indian Ocean West Pacific South Pacific Central Pacific East Pacific Atlantic



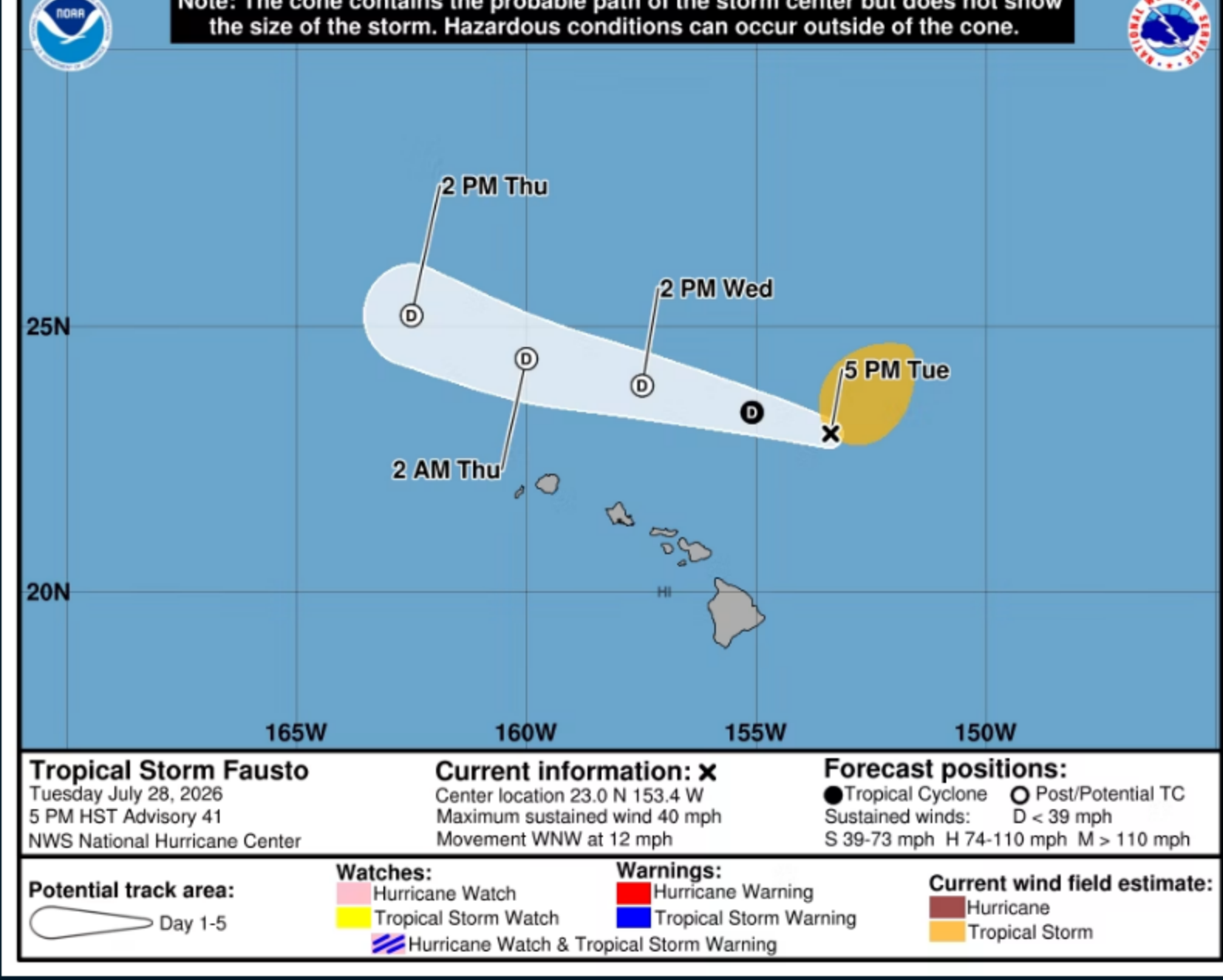
Tropical Storm FAUSTO

Tropical Storm Fausto Advisory Number 41
NWS Central Pacific Hurricane Center Honolulu HI EP06
Issued by NWS National Hurricane Center Miami FL
500 PM HST Tue Jul 28 2026

...FAUSTO CONTINUES TO GRADUALLY WEAKEN BUT SIGNIFICANT
EXPECTED FOR THE HAWAIIAN ISLANDS OVER THE NEXT FEW DAY

SUMMARY OF 500 PM HST...0300 UTC...INFORMATION

LOCATION...23.0N 153.4W
ABOUT 255 MI...405 KM NNE OF HILO HAWAII
ABOUT 310 MI...500 KM ENE OF HONOLULU HAWAII
MAXIMUM SUSTAINED WINDS...40 MPH...65 KM/H
PRESENT MOVEMENT...WNW OR 285 DEGREES AT 12 MPH...19 KM
MINIMUM CENTRAL PRESSURE...1006 MB...29.71 INCHES



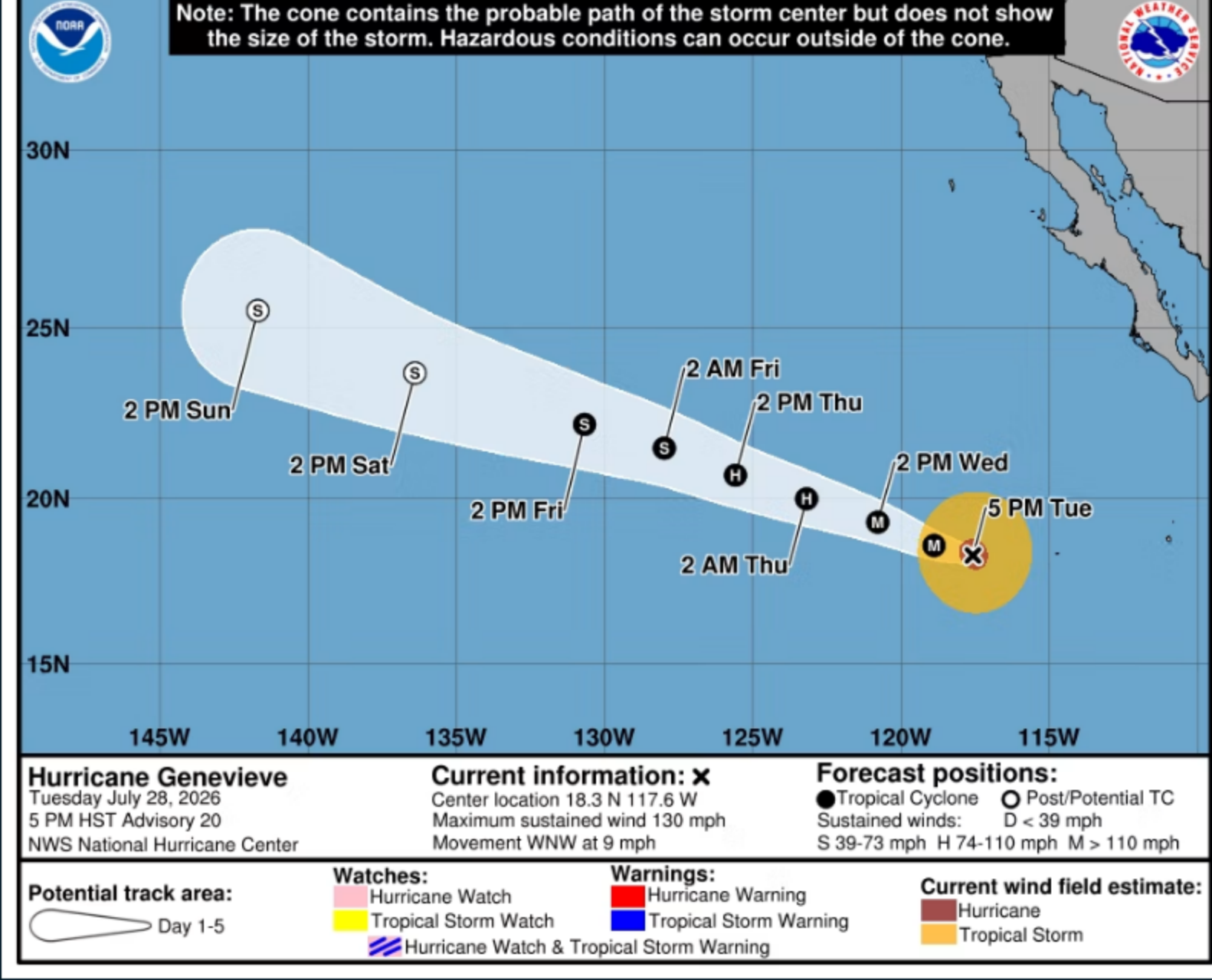
Hurricane GENEVIEVE

Hurricane Genevieve Advisory Number 20
NWS National Hurricane Center Miami FL EP072026
500 PM HST Tue Jul 28 2026

...MAJOR HURRICANE GENEVIEVE MOVING WEST-NORTHWESTWARD.

SUMMARY OF 500 PM HST...0300 UTC...INFORMATION

LOCATION...18.3N 117.6W
ABOUT 590 MI...950 KM WSW OF THE SOUTHERN TIP OF BAJA C
MAXIMUM SUSTAINED WINDS...130 MPH...215 KM/H
PRESENT MOVEMENT...WNW OR 290 DEGREES AT 9 MPH...15 KM/
MINIMUM CENTRAL PRESSURE...948 MB...28.00 INCHES



Typhoon DOLPHIN

1. TYPHOON 12W (DOLPHIN) WARNING NR 010
01 ACTIVE TROPICAL CYCLONE IN NORTHWEST PAC
MAX SUSTAINED WINDS BASED ON ONE-MINUTE AVERAGE
WIND RADII VALID OVER OPEN WATER ONLY

WARNING POSITION:
29000Z -- NEAR 14.5N 168.4E
MOVEMENT PAST SIX HOURS - 300 DEGREES AT 08 KTS
POSITION ACCURATE TO WITHIN 020 NM
POSITION BASED ON EYE FIXED BY SATELLITE

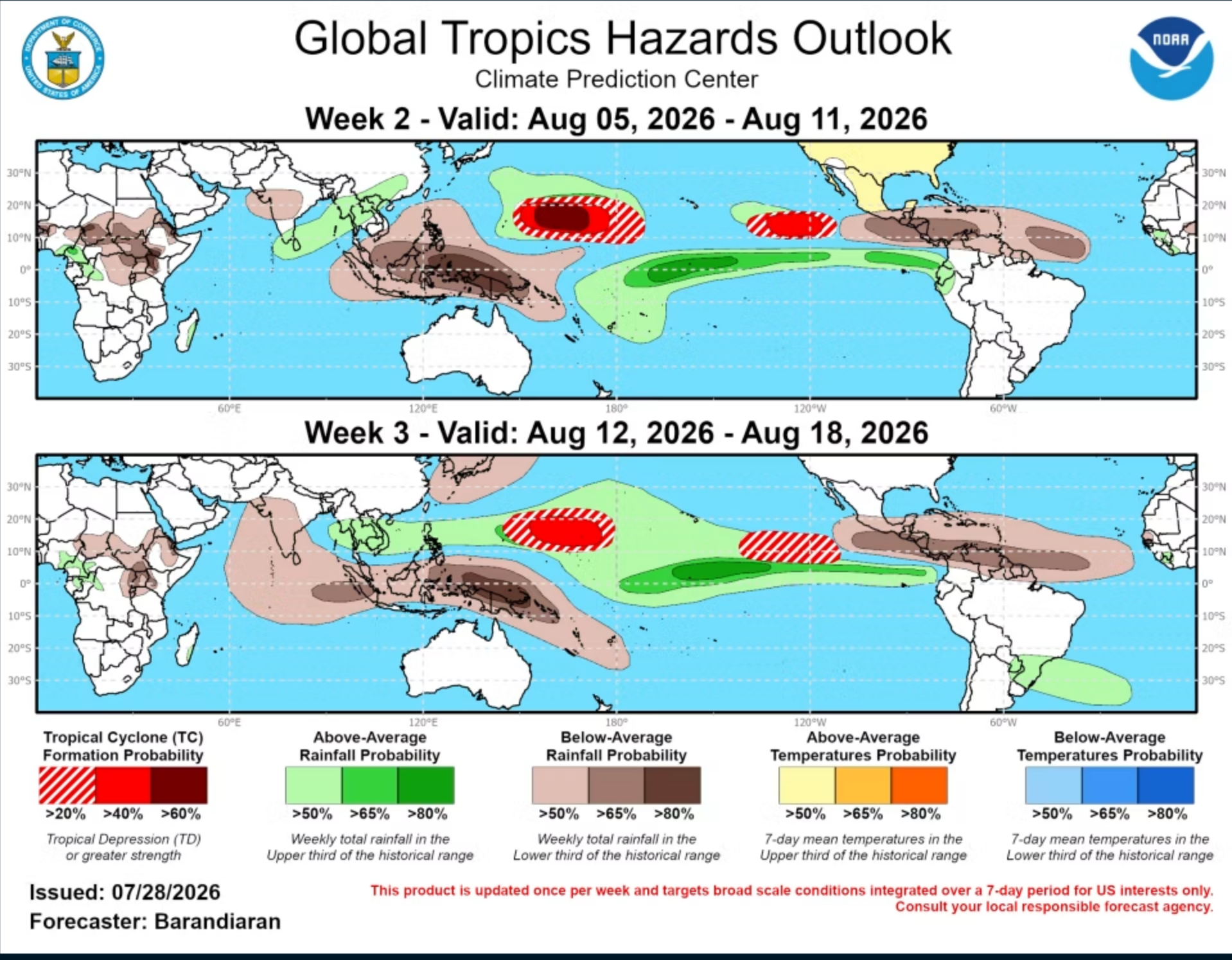
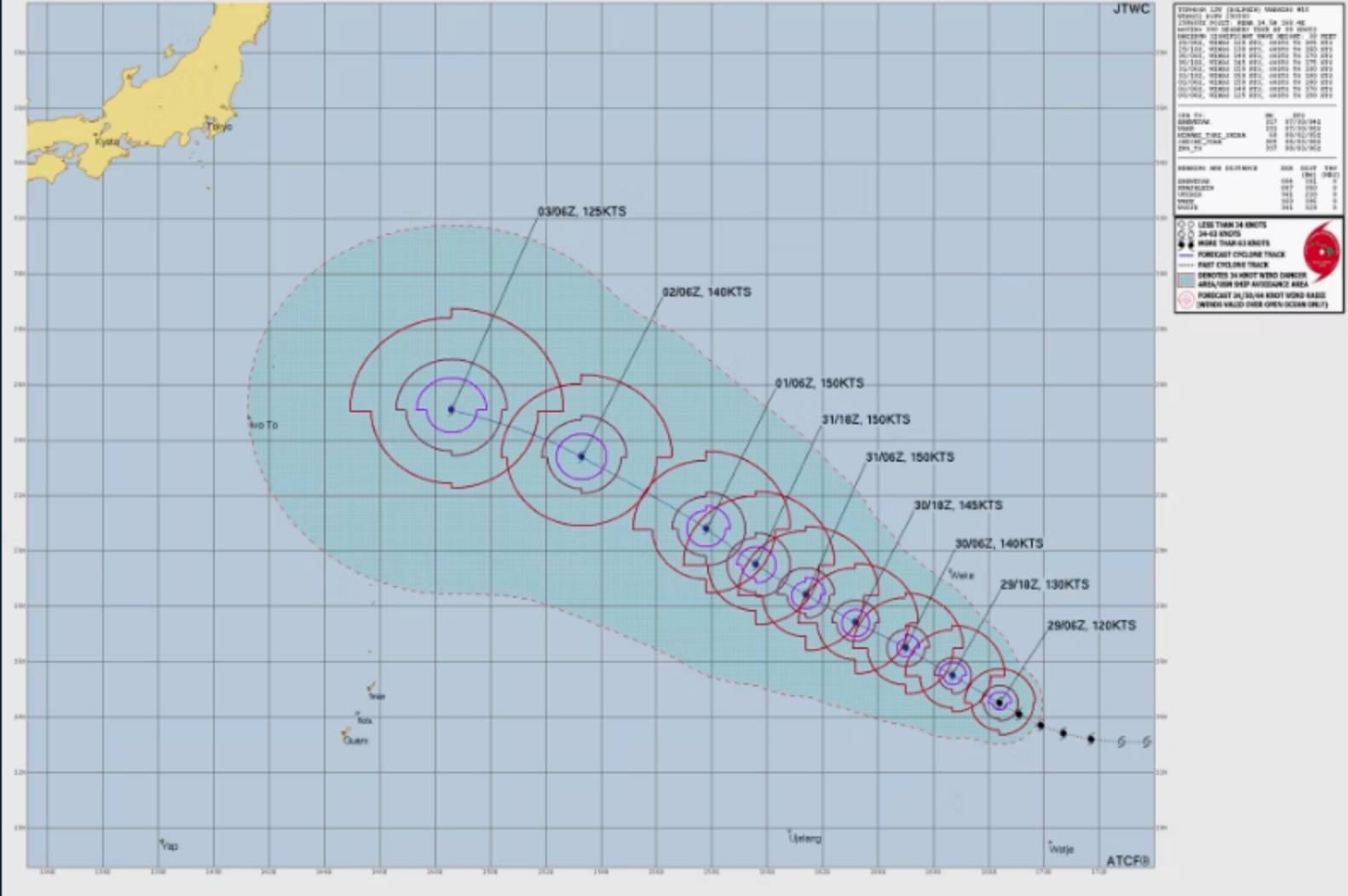
PRESENT WIND DISTRIBUTION:
MAX SUSTAINED WINDS - 120 KT, GUSTS 145 KT
WIND RADII VALID OVER OPEN WATER ONLY

RADIUS OF 064 KT WINDS - 025 NM NORTHEAST QUADRANT
015 NM SOUTHEAST QUADRANT
015 NM SOUTHWEST QUADRANT
025 NM NORTHWEST QUADRANT

RADIUS OF 050 KT WINDS - 040 NM NORTHEAST QUADRANT
040 NM SOUTHEAST QUADRANT
035 NM SOUTHWEST QUADRANT
040 NM NORTHWEST QUADRANT

RADIUS OF 034 KT WINDS - 075 NM NORTHEAST QUADRANT
070 NM SOUTHWEST QUADRANT
060 NM SOUTHWEST QUADRANT
075 NM NORTHWEST QUADRANT

REPEAT POSIT: 14.5N 168.4E



Graphic provided by Climate Prediction Center

See my new article: [How I Met Hurricane Camille](#)

What Is a Hurricane?

A hurricane (or typhoon, or severe tropical cyclone), the strongest storm on Earth, is a cyclonic (rotary) storm that derives its energy from cloud formation and rainfall, unlike frontal cyclones that derive their power from a temperature gradient.

A hurricane begins as a tropical depression with a sustained wind speed of less than 39 mph (35 knots; 63 km/hr). As the system strengthens, it becomes a tropical storm with winds from 39 to 73 mph (35-63 knots; 63-118 km/hr). Tropical storms are named in the Atlantic, East, Central and Northwest Pacific, in the South Indian Ocean, and in the Arabian Sea. When the winds are sustained (based on a one-minute average) at 74 mph (64 knots; 119 km/hr), the storm becomes: In the Atlantic Ocean, East Pacific, Central Pacific (east of the International Dateline) and Southeast Pacific (east of 160°E) a **Hurricane**; in the Northwest Pacific (west of the International Dateline) a **Typhoon**; in the Southwest Pacific (west of 160°E) and Southeast Indian Ocean (east of 90°E) a **Severe Tropical Cyclone**; in the North Indian Ocean a **Severe Cyclonic Storm**; and in the Southwest Indian Ocean (west of 90°E) a **Tropical Cyclone**.

The Saffir-Simpson Hurricane Scale

Category 1 – 64-82 knots (74-95 mph; 119-153 km/h). Damage is limited to foliage, signage, unanchored boats and mobile homes. There is no significant damage to buildings. The main threat to life and property may be flooding from heavy rains.

Category 2 – 83-95 knots (96-110 mph; 154-177 km/h). Roof damage to buildings. Doors and windows damaged. Mobile homes severely damaged. Piers damaged by storm surge. Some trees blown down, more extensive limb damage.

Category 3 – 96-112 knots (111-129 mph; 178-208 km/h). Major Hurricane. Structural damage to some buildings. Mobile homes are completely destroyed. Roof damage is common. Storm surge begins to cause significant damage in beaches and harbors, with small buildings destroyed.

Category 4 – 113-136 knots (130-156 mph; 209-251 km/h). Structural failure of some buildings. Complete roof failures on many buildings. Extreme storm surge damage and flooding. Severe coastal erosion, with permanent changes to the coastal landscape not unheard of. Hurricane force winds extend well inland.

Category 5 – 137+ knots (157+ mph; 252+ km/h). Complete roof failure on most buildings. Many buildings destroyed, or structurally damaged beyond repair. Catastrophic storm surge damage. In the Northwest Pacific, a typhoon that reaches 150 mph (241 km/hr) is called a Super Typhoon.

Category	SAFFIR-SIMPSON SCALE			Damage
	Knots	MPH	KM/H	
1	64-82	74-95	119-153	Minimal
2	83-95	96-110	154-177	Moderate
3	96-112	111-129	178-208	Extensive
4	113-136	130-156	209-251	Extreme
Super Typhoon	130+	150+	241+	Catastrophic
5	137+	157+	252+	Catastrophic

Storm Surge

Historically, storm surge is the primary killer in hurricanes. The exact storm surge in any given area will be determined by how quickly the water depth increases offshore. In deep-water environments, such as the Hawaiian islands, storm surge will be enhanced by the rapidly decreasing ocean depth as the wind-driven surge approaches the coast. The peak storm surge is on the right-front quadrant (left-front in the Southern Hemisphere) of the eyewall at landfall, where on-shore winds are the strongest, and at the leading edge of the eyewall. Contrary to a popular myth, the storm surge is entirely wind-driven water—it is not caused by the low pressure of the eye. Another factor in the severity of the storm surge is tide. Obviously, an 18-foot storm surge at high tide is that much worse than an 18-foot surge at low tide.

Tropical Cyclone Formation

By Jonathan Edwards

Tropical Cyclone Genesis is the technical term for the process of storm formation that leads ultimately to what are called hurricanes, typhoons, or tropical cyclones in various parts of the world.

This occurs when, in the Northern Hemisphere, the Intertropical Convergence Zone, or ITCZ, shifts northward out of the doldrums and atmospheric conditions become favorable for tropical cyclone formation after about the middle of May.

A series of low-pressure ripples develops within the ITCZ. These are known as tropical waves and progress from east to west. In the late season, they typically shift their movement toward the west-northwest, or even northwest, after crossing 45° or 50° W longitude.

These tropical waves, ideally embedded in the deep-layer easterly flow, contain a northeast wind shift. This is typically referred to as a “convergence”, where lines of equal atmospheric pressure are pressed together between the high-pressure ridge to the north and the developing low-pressure system. The divergence that results ahead of the convergence zone gives us a northeasterly wind as the axis of the tropical wave approaches. Gusts up to 25 mph may occur. Sometimes there can be gusts to tropical storm force in stronger waves. There can be next to no weather associated with these waves, and they may pass virtually unnoticed. More typically, there are bands of disturbed weather riding the axis of the wave. [Read more ...](#)